Wenhao Peng

RF Acoustic Wave Technology

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Biography

Wenhao Peng is a Staff Electrical Engineer at Skyworks Solutions Inc specializing in RF acoustic wave filter technology, where he develops resonator models and simulation methods that support filter design across BAW and SAW technologies. His published modeling work connects thermodynamics and mechanical vibrations with resonator and filter design. In ferroelectric BAW technology, he developed a physics based equivalent circuit model that bridges nonlinear ferroelectric thermodynamics and resonator DC bias dependent response. In piezoelectric BAW technology, he derived an energy based expression for electromechanical coupling as the ratio of converted electrical energy to supplied elastic energy and proved by modal analysis its consistency with the coefficient determined from the series and parallel resonance frequencies when the mode is sufficiently isolated from other modes, making modal field distributions useful for evaluating layer stacks and device performance at mm-Waves. Experimentally, he demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBAR operating above 50 GHz, a quad band switchless filter bank, an acoustic phase inversion switch, and an intrinsically switchable balanced filter. At Skyworks, he is developing modal and transfer matrix approaches to accelerate established finite element models for both BAW and SAW and integrate them into the filter design workflow. He received the Ph.D. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, in 2026, the M.S. degrees in mechanical engineering in 2024 and in electrical and computer engineering in 2019, and two B.S.E. degrees in 2018. He has published in IEEE T-MTT, IEEE MWTL, IEEE MWCL, Applied Physics Letters, has presented as the first author at IEEE/MTT-S International Microwave Symposium in 2021, 2024, and 2025, and serves as a peer reviewer for leading journals in IEEE and APS.

Original Contributions

Modal analysis for linear resonator response

Derived an energy based expression for the electromechanical coupling coefficient (k_t^2) as the ratio of converted electrical energy to supplied elastic energy, and proved its consistency with the coefficient determined from the series and parallel resonance frequencies. The expression was derived through modal analysis of the open circuit eigenmodes of an FBAR. The result makes acoustic fields at resonance quantitatively interpretable, showing the contribution of each layer to resonator coupling, and enables direct evaluation of stack performance, reducing reliance on iterative optimization against simulated admittance. This relation was applied in both mm-Wave FBAR papers, with the derivation presented at IMS 2024 and IMS 2025.

Physics based models for nonlinear resonator response

Adapted finite element interpolation methods to bridge a thermodynamic description of ferroelectric materials and a compact circuit representation of acoustic resonance. Developed the first equivalent circuit model of ferroelectric driven BAW resonators based on thermodynamics of ferroelectric materials which is validated with measured response across DC bias. First author of the corresponding IEEE T-MTT 2022 and IMS 2021 papers.

mm-Wave scaling by periodically poled films — Demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBAR operating at 56 GHz. The modal analysis result guided electrode and layer stack selection for a second generation 36 GHz device. First author of the corresponding IMS 2024 and IMS 2025 papers.

Switchless RF frontends — Demonstrated the first switchless quad band filter bank based on ferroelectric BST FBARs, the first intrinsically switchable balanced ferroelectric FBAR filter, and the first RF acoustic phase inversion switch based on poled ferroelectrics.

Experience

Feb 2026–Present — **Staff Electrical Engineer**, *Skyworks Solutions Inc*, Irvine, CA.

- Provide device modeling and simulation support for BAW and SAW filter designs.
- Develop modal and transfer matrix methods to accelerate established finite element models for both BAW and SAW and integrate them into the filter design workflow.
- Diagnose resonator and filter performance issues through mass production using equivalent circuit and multiphysics models.

Sep 2019–Dec 2025 — **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.
Advisor: Prof. Amir Mortazawi

mm-Wave acoustic resonators based on periodically poled III-nitride thin films

- Demonstrated the first periodically poled AlN/ScAlN/AlN trilayer FBARs at 56 GHz, followed by a second generation 36 GHz resonator with a redesigned electrode and layer stack.
- Derived an energy based expression for electromechanical coupling from the open circuit eigenmodes of an FBAR, defining the coupling coefficient as the ratio of converted electrical energy to supplied elastic energy and proving its consistency with the value determined from the series and parallel resonance frequencies. The expression enabled direct evaluation of layer contributions and guided electrode and layer stack selection for the mm-Wave FBAR work.
- Advanced the nanofabrication and pulsed-field polarization switching processes of MBE grown AlN/ScAlN multilayer structures.
- Performed mm-Wave characterization to 67 GHz with on-wafer de-embedding structures validated in ADS Momentum and Ansys HFSS.

Reconfigurable RF frontends based on intrinsically switchable BST acoustic resonators

- Developed the first equivalent circuit model of ferroelectric driven BAW resonators based on ferroelectric thermodynamics. Adapted finite element interpolation methods to bridge the material thermodynamic description and a compact circuit representation of acoustic resonance. Validated the model by direct comparison with measurements across DC bias.
- Demonstrated the first switchless quad band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
- Demonstrated the first RF acoustic phase inversion switch based on poled ferroelectric stacked crystal filters.
- Synthesized filters using the image parameter method, applying Mason and mBVD resonator models, and completed cleanroom fabrication and DC bias dependent characterization.

Education

2026 — **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Dissertation: *Layered Ferroelectric Millimeter Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization.* Advisor: Prof. Amir Mortazawi.

2024 — **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.
Dynamics and vibrations

2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Integrated circuits and VLSI

2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.
Summa Cum Laude

2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*, Shanghai, China.
Dual degree with U-M

Technical Skills

Devices & materials	BAW/FBAR resonator and filter design; thin film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
Modeling	Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); structural vibration analysis
Fabrication	Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE-grown films; pulsed-field poling
Measurement	On-wafer S-parameter characterization to 67 GHz; GSG probing; VNA calibration and de-embedding; DC bias dependent RF characterization; material parameter extraction
Tools	ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence
AI-assisted work	LLM-assisted Python development, code review, numerical workflow automation, and technical documentation

Journal Publications

2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.

2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.

2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.

2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.

2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, "A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR," *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*

2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, "A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR," *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*

2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Physics Based Modeling of Electrostriction Based BAW Resonators," *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*

2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, "An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz," *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.

2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, "Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication," *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review Contributions

51 reviews **IEEE:** J. Microelectromechanical Systems (9), Microwave and Wireless Technology Letters (6), Electron Device Letters (4), J. Microwaves (2). **APS:** Physical Review Letters (5), Physical Review Applied (3), Physical Review Materials (2), Physical Review A (1), Physical Review B (1). **Other:** Physics of Fluids (1), Micromachines (10), Photonics (2), Sensors (2), Microwave (MDPI) (2), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Math Methods in ME; Advanced Solid State Microwave Circuits; Electromagnetics Theory; Introduction to MEMS; Monolithic Amplifier Circuits; Analog Integrated Circuits; VLSI Design; Computer Architecture; Digital Communication Signals and Systems; Probabilistic Methods in Engineering